STADE-CDNet: Spatial–Temporal Attention With Difference Enhancement-Based Network for Remote Sensing Image Change Detection

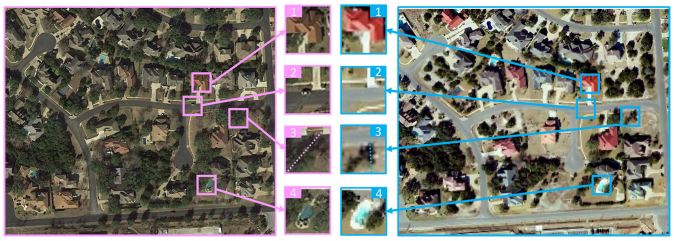
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***Abstract*— High-resolution remote sensing (RS) image change detection (CD) focuses on ground surface changes. It has wide applications, including territorial spatial planning, urban region detection, and military operations. However, class imbalance and pseudochanges are caused by the unchanged areas far outnumbering the changed areas and lighting changes. To address these problems, we propose spatial–temporal attention with a difference enhancement-based network (STADE-CDNet). In STADE-CDNet, a CD difference enhancement module (CDDM) is proposed to extract important features from the difference map to detect changed regions. This module enhances the network with differential feature attributes through the training layer, improving the network's learning ability and reducing the imbalance problem. A temporal memory module (TMM) is designed to extract temporal and spatial information. Inspired by the self-attention mechanism of the transformer, we propose a transformer and TMM (TTMM). Four encoding layers are designed to detect the semantic information from high to low levels of the multitemporal image pairs. The fusion and parallelism of multivariate data are achieved through collaborative modeling of deep learning and CD, compensating for the need for excessive human intervention in traditional algorithms. We evaluate our approach in two different datasets [LEVIR building CD (LEVIR-CD) and deeply supervised image fusion network for change detection (DSIFN-CD)]. Promising quantitative and qualitative results show that the STADE-CDNet can improve accuracy. In particular, the proposed CDDM significantly reduces false positive detection, with $F1$ scores at least 1.97% and 2.1% higher than other methods in the case of the LEVIR-CD and DSIFN-CD datasets, respectively. Our code is available at https://github.com/LiLisaZhi/STADE-CDNet.**

***Index Terms*— Change detection (CD) difference enhancement, multitemporal image pairs, temporal memory, transformer.**

I. Introduction

AS SATELLITE technology develops, remote sensing (RS) images have a wider variety, such as medium-resolution, high-resolution, and other RS images; hence, RS images are becoming clearer and providing more detailed information [1], [2], [3], [4], [5], [6]. RS change detection (CD) image processing detects changed areas on the ground surface through qualitative and quantitative analysis in different dimensions. It has wide and deep applications in territorial spatial planning, urban region changes, disaster monitoring, and military and agriculture operations [6], [7], [8], [9], [10], [11]. The most critical changes in high-resolution RS images are building growth and changed regions in land cover (including building changes). These cases cause two main problems.



(a) (b)

Fig. 1. Example of two images taken at different times. (a) Prechange image. (b) Postchange image.

First, as the prechange/postchange images in Fig. 1 show (from the LEVIR building CD (LEVIR-CD) dataset), typical pseudochanges problems can be categorized as (1) and (4) changed item color, (2) temporary object presence, and (3) various shade shapes. These pseudochange problems [12], [13], [14], [15] lead to shadows and projection differences in unchanged areas being mistakenly detected as changed areas. Moreover, these problems may invalidate battlefield information, leading to serious consequences. Second, the RS CD (RSI-CD) datasets usually have more unchanged areas than changed areas, leading to class imbalance problems. This issue can cause missing small changed regions, such as the color of a house roof, which is common in most CD datasets. This leads to a significant amount of missed changed information and reduced detection accuracy.

Therefore, the pseudochanges and class imbalance problems are essential challenges and worth solving. Consequently, we develop a novel approach to resolve the pseudochanges and class imbalance problems in high-resolution RSI-CD.

A. Related Works

Many studies have focused on solving pseudochanges and class imbalance problems. Traditional CD methods are

Manuscript received 5 September 2023; revised 9 January 2024; accepted 29 January 2024. Date of publication 20 February 2024; date of current version 29 February 2024. This work was supported in part by the Natural Science Foundation of Sichuan Province of China under Grant 2022NSFSC40574 and Grant 2023NSFSC0508 and in part by the National Natural Science Foundation of China under Grant 61775030 and Grant 61571096. *(Corresponding authors: Zhenming Peng; Junhai Luo.)*

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Digital Object Identifier 10.1109/TGRS.2024.3367948

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usually based on object-level or pixel-level detection. Pixel-level detection approaches typically detect changed regions by comparing pixel values. Kondmann et al. [16] used sibling regression methods to transform pixel modeling into linear combinations of domains to reduce pseudochanges and class imbalance and ultimately predict differences in images. Zhao et al. [17] reduced pseudochanges by extracting spatial and frequency domain features, ultimately improving detection accuracy. Object-level approaches consider the spectral information of the spatial and overall original image and then detect changed areas. For example, Wu et al. [18] finally detected the changed areas by segmentation, figure construction, and change measurement using both object and common boundary definition weights. In object-level detection, the pseudochanges and class imbalance problems were also addressed. However, traditional CD methods are limited by the assumption that they process the same class of image data. Thus, they do not overcome the challenges of fusion, parallelism, and complementarity of multiple RS data.

With technological innovation, deep learning has been successfully applied to RSI-CD tasks [19], [20], [21], [22]. The problems faced by traditional pixel-level and object-level detection methods can be solved with higher accuracy and robustness by deep learning methods. As a result, a series of neural networks have been developed, which have demonstrated superior image classification and CD results [23], [24].

Nevertheless, scholars find that adopting convolutional neural networks in extracting information leads to long running times and challenges, such as gradient disappearance as the convolutional layers deepen. Therefore, effectively using contextual information while efficiently detecting valid information has become a hot research topic [25], [26], [27]. In response to these challenges, researchers have proposed transformers in the field of natural language processing. The transformer, which was initially proposed for natural language processing, is a deep learning framework comprising an encoder and a decoder. The encoder module handles the input data and extracts useful features, while the decoder module generates the output by using these features. Furthermore, the transformer extracts the interest contextual information through a self-attention mechanism model [28].

Therefore, by accelerating real-time contextual feature extraction, the transformer enhances the efficiency of the model. In addition, it also contributes to increased algorithmic detection accuracy by efficiently identifying valuable information, because each block of the model has its own set of features of interest. In RS images, transformers have often been used in fields, such as image classification and image recovery [29], [30], and have performed well in classification and detecting critical information. In the context of CD, transformers have been applied to CD. The encoder–decoder structure of the transformer network furthers the extraction and fusion of spatial information, enabling the effective detection of changed areas in multitemporal images. More specifically, in the CD task, the transformer network takes the multitemporal images as input. The encoder captures the spatial features and the vital contextual information of the images, which is crucial for accurate changed areas detection. After that, the decoder uses this encoded information to generate a change map, highlighting the regions where significant alterations have occurred. By utilizing the transformer network's attention mechanisms and self-attention mechanisms, the model can effectively learn complex spatial patterns. This empowers the network to improve the accuracy and robustness of CD, making it a promising approach in this field of research [31].

However, despite the potential of the transformer network to improve accuracy and robustness in CD, exploring the transformer network, reducing the number of operations on the model, and enhancing the algorithm's generalization capability remain challenging tasks.

B. Motivation

CD data typically consists of spectral vectors in both temporal and spatial dimensions, and developing relationships between different spatial and temporal locations can enhance the effectiveness of CD methods [32]. This also implies that CD is associated with sequential data; therefore, it is necessary to consider temporal dynamics. Long short-term memory (LSTM) networks [33] reduce the gradient disappearance problem in dealing with long-term dependencies compared with previous recurrent neural networks (RNNs) [34]. Scholars have applied LSTM to CD in RS images by integrating a deep neural network-based recommendation system (ReCNN) into a convolutional architecture to exploit spatial and temporal features in an end-to-end framework. In particular, changed information is detected by acquiring the corresponding pixels of multitemporal images and modeling the temporal relationships through multitemporal RS images [35]. To exploit the temporal feature information of LSTMs, scholars have combined LSTM-UNet to form a recurrent network that produces a more productive feature representation. LSTM-UNet extracts both spatial and temporal information during the learning process. In the network, the LSTM is placed as a convolutional layer on top of each coding level of a UNet-like deep architecture, so that changed spatial and temporal information is detected [36]. In addition, recent works, such as the temporal difference-guided network and the multiscale difference-changed feature fusion network, have also been proposed for CD [37], [38]. These models aim to further improve the accuracy and efficiency of CD by incorporating specific techniques for tackling temporal dynamics and integrating multiscale feature fusion capabilities. However, these models have limitations, such as limited ability, to obtain long-range dependencies and difficulties in handling complex spatial patterns, which affect the overall performance and accuracy of the model.

In RSI-CD, a difference map of multitemporal images clearly shows the changed areas, making it easier to improve the method to obtain more realistic changed areas [39]. Moreover, the detection of difference maps further enables accurate identification and precise localization of the changed areas. However, if we adopt pixel subtraction, the differences in local image features are ignored. From the above analysis, we can see that the difference map can reflect the difference between the input multitemporal images and improve the learning

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ability of the network. Therefore, it is necessary to learn the difference map in CD.

Driven by the above analysis, we develop and evaluate a deep learning framework that proposes spatial–temporal attention with a difference enhancement-based network (STADE-CDNet). In STADE-CDNet, we design the transformer encoder with four encoding layers to detect the semantic information from high to low levels of the multitemporal image to solve the pseudochanges problem. Here, we put the temporal memory module (TMM) behind the transformer encoding, not only to mitigate gradient decay during backpropagation but also to extract spatial and temporal information in learning to generate cost-effective feature representations. Moreover, we propose a CD difference enhancement module to improve the learning ability of the network by learning spatial attention and channel attention of the difference maps simultaneously, reducing the CD class imbalance problem. In summary, the contributions made in this article are as follows.

1) We propose the STADE-CDNet. In this network, we propose two main modules, the CD difference enhancement module (CDDM) and TTMM, to reduce the pseudovariation and class imbalance problems in high RSI-CD.
2) The CDDM is further explored under the framework. This module learns the spatial attention and channel attention of the difference map, then generates an enhanced difference feature map, and, finally, improves the learning capability of the network.
3) A transformer and TMM (TTMM) is proposed. First, the TMM is proposed to extract temporal information in high-resolution RS image pairs to accurately detect changed regions. Second, we extract the temporal and spatial information of the images by using the TTMM to model the extracted features.

The remaining sections of this article are structured as follows. Section II describes the method along with the implementation details. Section III presents the benchmark datasets, the evaluation metrics, the ablation experiments, the experimental results, and the discussion. Section IV concludes this article.

II. Methodology

In this section, the structure of STADE-CDNet is introduced first. Subsequently, we introduce a detailed explanation of the TMM. Finally, the CDDM is elaborated upon.

A. Overview

The overview of STADE-CDNet is illustrated in Fig. 2. As depicted in the figure, STADE-CDNet consists of three main components: the backbone, the TMM, and the CDDM.

To date, Scholars have applied transformers to process images. Important information is extracted, and irrelevant details are discarded using self-attention, thus improved outcomes [29]. Applying a transformer to high-resolution RSI-CD is also a hot topic of current research [40]. Therefore, a network framework is developed by inputting multitemporal RS image pairs to the transformer combined with CNN. The CNN is used to extract high-level semantic information from the difference map, and then, the transformer is used to encode and decode the changed areas of interest. Finally, the changed areas are obtained [31]. However, we also know that the pseudochanges problem is one of the most difficult aspects of CD, as the low-level semantic information contains information, such as color, geometry, and texture in the image, which contributes to pseudochanges. Chen et al. [31] ignored the low-level semantic information, which led to pseudochanges and inaccurate CD; therefore, we should detect as much semantic information as possible from high to low levels.

In addition, considering that CD datasets are usually composed of spectral vectors in both temporal and spatial dimensions, our proposed TMM method aims to extract temporal information from multitemporal RS images. By integrating this temporal information, we can enhance the accuracy and reliability of CD.

However, when capturing temporal relationships in CD datasets using traditional approaches, LSTM becomes slightly inefficient. This is because, in the case of high-dimensional data, such as RS images, the number of parameters is much higher than that in natural language processing, resulting in a high training cost. Consequently, training the LSTM model becomes computationally expensive. Moreover, if only the temporal information of the CD images is detected, there is a risk of overlooking important spatial information present in the images, leading to a decrease in detection accuracy.

To address these challenges, we propose the use of the TTMM inspired by the literature [36]. This module replaces the conventional LSTM with a TMM that passes multitemporal RS images into this model. The transformer serves as the network backbone and is divided into four layers in the encoder, allowing for semantic information extraction from high to low levels in the images. The TMM is positioned after the transformer encoder, as shown in Fig. 2.

Furthermore, we propose a CDDM to extract more useful information from the difference map in the RS images. This module focuses on the difference map derived from two input feature maps, further improving network performance. In our experiments, we compared various modules and found that our proposed module outperformed other methods. Therefore, incorporating the TTMM and CDDM modules into our framework helps us to better extract temporal and spatial information in CD datasets, ultimately solving imbalance and pseudochanges problems.

In summary, we propose a STADE-CD network. This network utilizes the transformer as its backbone to extract information at different levels in RS images. The TTMM is employed to capture both temporal and spatial information. In addition, the CDDM is introduced to analyze the information within the difference map using deep learning techniques.

B. Temporal Memory Module

RNNs are deep learning algorithms that have been widely used in processing temporal data. However, the original RNN in most applications cannot obtain desirable results due to gradient disappearance [35]. To solve this problem, they propose LSTM. This problem is solved by introducing memory

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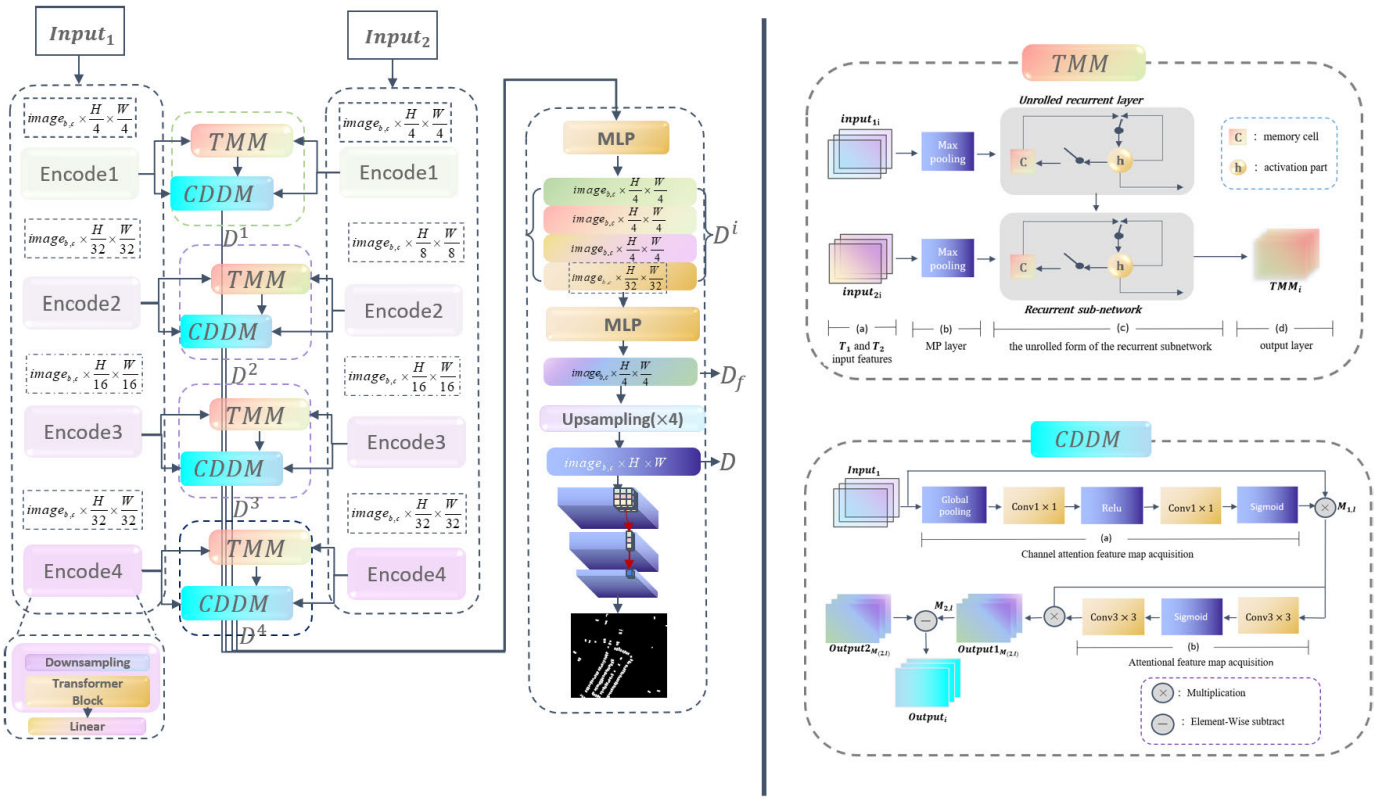


Fig. 2. Overall framework of STADE-CDNet.

cells (often referred to as "cell states"), which prevent gradient values from disappearing during longer data transfers. Thus, they maintain training stability during deep learning [41].

LSTM can remember long-term information while solving the dependency problem very well. In general, RNNs obtain information on the temporal dimension by using a single hyperbolic tangent layer at each time point. In contrast, LSTMs extract temporal information by using four gating units. The following equations show the role of the individual gates: forget gate ($\mathbf{f}_t$), input gate ($\mathbf{i}_t$), unit gate ($\tilde{C}_t$), and output gate ($\mathbf{O}_t$) [41]:

$$f_t = \sigma\left(W_f \cdot (X_t, H_{t-1})\right) \tag{1}$$

where σ is the sigmoid function, W_f is weight matrices employed for the forget gating unit, and H_{t-1} is the hidden states

$$i_t = \sigma(W_i \cdot (X_t, H_{t-1})) \tag{2}$$

where W_i is weight matrices employed for the gating unit and the input gate determines the extent to which candidate values can be stored in the current cell state.

The memory cells C_t are updated by partially discarding the present memory contents and adding new contents of the memory cells $\tilde{C}_t$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{3}$$

where $\odot$ is an elementwise multiplication. The new memory contents are

$$\tilde{C}_t = \tanh(W_c \cdot (X_t, H_{t-1})) \tag{4}$$

where W_c is the input-memory weight matrix and tanh is the hyperbolic tangent function. In (4), the cell gate utilizes the hyperbolic tangent function to regulate the data a produce possible candidate cell state value.

The output gate further filters the information determining the outcome of the network

$$O_t = \sigma(\mathbf{W}_o \cdot (X_t, H_{t-1})) \tag{5}$$

where $\mathbf{W}_o$ is the corresponding weight matrix.

Ultimately, the cell state is passed through a hyperbolic tangent function and multiplied by the output gate to generate the hidden state of the current time step t, formulated as follows:

$$H_t = O_t * \tanh(C_t) \tag{6}$$

where O_t is the output gates that control the amount of memory content exposure.

Inspired by Chen et al. [37] and Luo et al. [38], we denote the feature maps from the two images at level i of the transformer as input_{1i} and input_{2i}, as shown in Fig. 3. Subsequently, input_{1i} and input_{2i} are fed into the time units T_1 and T_2 of the TMM separately. In particular, before input_{1i} is fed to T_1, the hidden state (H_0) of T_1 is initialized to zero, which indicates the clearing of the past state and the unknown of future information. As a result, the output of the time unit T_1 is considered as the hidden state of the T_2 to influence its output. Due to the temporal similarities between the features of the bitemporal images in the CD task, T_1 overlooks these similarities and focuses on highlighting the

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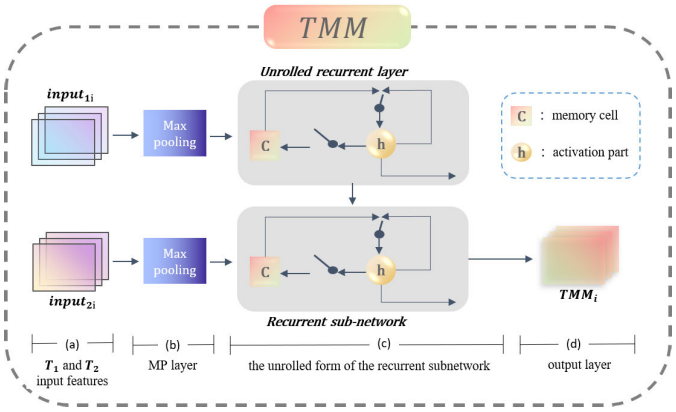


Fig. 3. TMM architecture figure.

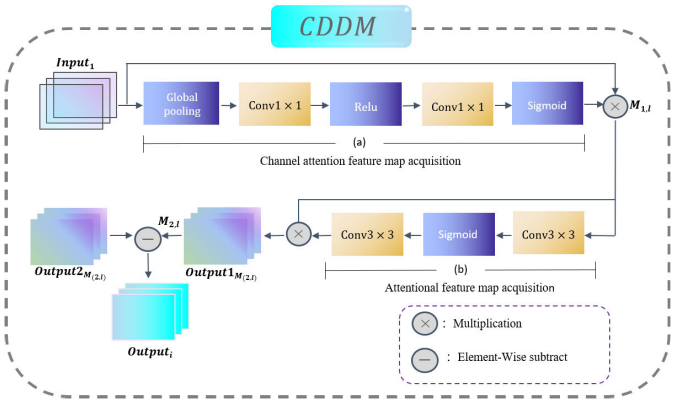


Fig. 4. CDDM.

differences between the two images. Consequently, the output (H_2) of T_1 corresponds to the identified change features and the formulated as follows:

$$D_i = \mathrm{LSTM}_i\left(\mathrm{input}_{1i}, \mathrm{input}_{2i}\right). \tag{7}$$

In the context of CD, LSTM exhibits the following limitations. Initially, it causes high training costs when dealing with high-dimensional data. Processing high-dimensional data, such as RS images, increases the number of model parameters, resulting in elevated training complexity and computational overhead. Moreover, LSTM primarily focuses on modeling time-series data, thereby disregarding spatial information within images. This oversimplification of spatial relationships and contextual information can negatively impact the accuracy of CD.

To address these issues, we have introduced a max-pooling (MP) layer before the LSTM to extract temporal information from high-resolution RS images and mitigate the abovementioned limitations' impact on CD. The incorporation of the MP layer provides two primary advantages.

First, MP reduces the dimensionality of the data, thereby decreasing the number of parameters in the LSTM model and alleviating training costs and computational complexity. This is particularly beneficial when dealing with high-dimensional data, such as RS images.

Subsequently, MP aids in capturing the most salient local features within the image, extracting crucial spatial information for CD. By introducing MP before the LSTM, we can better preserve the significant features within the image, consequently enhancing CD accuracy.

In conclusion, by incorporating an MP layer before the LSTM, we overcome several limitations in CD, including the high training costs associated with high-dimensional data and the disregarding of spatial information, as shown in Fig. 3. The formula is as follows:

$$\mathrm{TMM}_i = \sigma(\mathrm{MP}(D_i)) \tag{8}$$

where the operators MP(•) represent the MP layer.

The TMM updates the state of a cell gate by forgetting irrelevant features in the unit gate while retaining the valuable features of the unit state. The TMM carefully filters the flow information in the time dimension. It not only retains useful information in the time dimension well but also effectively mitigates the weight-sharing problem. The above analysis shows that the TMM mitigates the gradient disappearance problem during backpropagation. Furthermore, the TMM is effective in generating useful feature information by extracting spatial and temporal information in the learning process, thereby enhancing the learning ability of the network.

High-resolution RS image CD datasets usually consist of spectral vectors in both the temporal and spatial dimensions. Developing relationships between different spatial and temporal locations can enhance the performance of CD approaches. Therefore, we extract spatial and temporal information from multitemporal RS image change pairs to accurately detect changed regions.

C. Change Detection Difference Enhancement Module

Once the input RS images have been processed by the TTMM subnetwork, the spatial and temporal feature maps are obtained. These feature maps typically consist of both changed and unchanged regions, where the changed regions are particularly effective in detecting and highlighting small changed buildings. However, previous studies have generally overlooked the learning of changed information that differential changed features can effectively represent.

To handle this limitation, we propose a novel CDDM module designed specifically to learn different features. This module leverages the multilevel feature maps generated by the CDDM subnetwork to extract valuable information. The structure of the CDDM module is illustrated in Fig. 4.

In RSCD, the difference map plays a crucial role in both quantitative indicators and qualitative judgments of detecting changed areas. However, performing direct pixel-level subtraction hinders the network from learning relevant difference map information and leads to reduced detection accuracy. Hence, it is essential to incorporate the learning of the difference map in RSCD.

Initially, we pass the input feature maps through a global pooling unit. Second, complex correlations between channels are fit by two 1×1 convolutional layers, and the feature maps are downscaled and then upscaled, adding more nonlinear processing. Next, the feature maps are passed through the sigmoid layer to obtain a $1 \times 1 \times C$ channel attention mask.

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Finally, the channel attention mask is multiplied by the input feature map to obtain the channel attention feature map $M_{1,l}$, as shown in the following equation:

$$M_{1,l} = \text{Input}_1 \otimes \text{Sigmoid}\big(\text{Conv}_{1\times1}\big(\text{ReLU}\big(\text{Conv}_{1\times1}\big(G_p\big(\text{Input}_1\big)\big)\big)\big)\big). \quad (9)$$

To obtain the spatial attentional feature maps, a 3 × 3 convolution operation and a sigmoid operation are applied to $M_{1,l}$ to obtain the attentional feature maps: $M_{2,l}$. This is shown in the following equation:

$$\text{Output}_{1_{M_{2,l}}} = M_{1,l} \otimes \text{Conv}_{3\times3}\big(\text{Sigmoid}\big(\text{Conv}_{3\times3}\big(M_{1,l}\big)\big)\big). \quad (10)$$

The feature map $M_{2,l}$ contains the channel attention and spatial attention features of the input feature maps. Finally, in the output feature map, the enhanced multitemporal input feature map is subtracted to obtain the final difference-enhanced difference feature map, as shown in the following equation:

$$\text{Output}_i = \text{abs}\Big(\text{Output}_{1_{M_{2,l}}} - \text{Output}_{2_{M_{2,l}}}\Big). \quad (11)$$

In general, learning difference maps through the CDDM improves the overall network learning. This enhancement can be attributed to two key factors: increased accuracy and enhanced robustness.

On the one hand, the CDDM module enhances accuracy in change detection by introducing additional feature extraction and data processing mechanisms. It effectively captures the change information within the images and efficiently filters out noise and redundant information, resulting in more reliable results. This improvement enables the network to better discover subtle changes and produce more precise detection outcomes.

Besides, the CDDM module strengthens the robustness of the network by considering the resilience of the data. It can handle various types of changes, such as variations in lighting conditions, occlusions, and noise. Through appropriate data processing and modeling methods, the CDDM module adapts well to complex scenes, providing more stable and reliable CD results. This robustness ensures the network's ability to handle diverse and challenging scenarios, enhancing its overall performance.

In summary, the CDDM module improves the network's accuracy by better-capturing change information and filtering out noise and redundancy. In addition, it enhances the network's robustness by effectively handling different types of changes. Overall, learning difference maps through the CDDM module contributes to improved network learning and performance.

III. Experimental Results and Discussion

In this section, we present the benchmark datasets, the evaluation metrics used, and ablation studies. In the ablation experiments, we keep the learning rate and each parameter, such as the optimizer, unchanged to ensure fair comparisons. In addition, we compare the proposed algorithm with other advanced methods. Finally, we discuss the algorithms.



Fig. 5. LEVIR-CD dataset. (a) Prechange image. (b) Postchange image. (c) Ground truth.

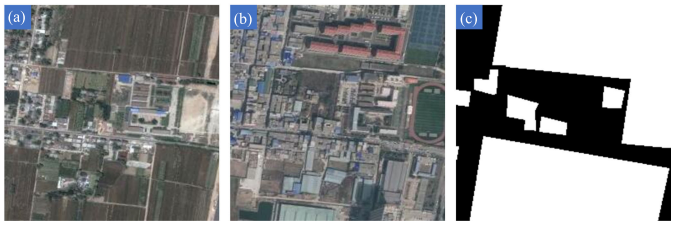


Fig. 6. DSIFN-CD dataset. (a) Prechange image. (b) Postchange image. (c) Ground truth.

A. Datasets and Implementation Details

The conducted experiments are based on two high-resolution RS image datasets: the high-resolution LEVIR-CD [42] and the deeply supervised image fusion network for change detection (DSIFN-CD) [43]. For all datasets, the main task is to detect both changed and unchanged regions.

LEVIR-CD is a new large-scale RS-built CD dataset. Multitemporal image pairs were collected over a period between 5 and 14 years. In this dataset, scholars were mostly concerned with building growth (changed from soil/grass/hardened ground or buildings under construction and new building areas) and building decline. Therefore, changed regions are buildings, and unchanged regions are nonbuildings. The LEVIR-CD dataset consists of 637 high-resolution (0.5 m/pixels) Google Earth (GE) image patch pairs, 1024 × 1024 pixels in size. We divided the dataset randomly into three parts as train/val/test sets of samples 445/10/128. The original image pairs and ground truth are shown in Fig. 5.

The DSIFN-CD dataset, which we obtained manually from GE, is a valuable resource for our research. This dataset includes five large high-resolution images, encompassing six cities in China: Beijing, Chengdu, Shenzhen, Chongqing, Wuhan, and Xi'an. These images offer a comprehensive representation of the urban landscape in different seasons. The DSIFN-CD dataset is meticulously annotated, enabling accurate detection of changed regions in the ground truth. This includes detecting the occurrence and disappearance of various land-cover objects, such as roads, buildings, agricultural land, and water bodies. These changes provide valuable insights into the dynamics and transformations of urban environments. For our experiments, we adopt the default train/val/test sets provided by the dataset authors. The dataset is divided into 445 training samples, ten validation samples, and 128 test samples, ensuring a robust evaluation of our model's performance. To provide a visual reference, we present the original image pair and corresponding ground truth in Fig. 6. These visuals offer a glimpse into the dataset and the changes that our model aims to detect and analyze.

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B. Optimization

The hyperparameters are similar for both datasets, using the AdamW optimizer. Learning rate is 9.5×10^{-6}. To ensure efficient training, we employ a batch size of 8 for both the LEVIR-CD dataset and the DSIFN-CD dataset. Our model is implemented in PyTorch and trained on a single NVIDIA TITAN RTX with 24G of GPU memory.

To initiate the training process, we begin by randomly initializing the network. Subsequently, we train the model using a cross-entropy (CE) loss function, with weights that decay to 0.05 and beta values set to (0.9, 0.999). This approach enables us to effectively optimize the model's performance and achieve accurate results.

C. Quantitative Evaluation Metrics

Five different evaluation metrics are used to evaluate quality: precision, recall, $F1$, intersection-over-union (IoU), and kappa. Their values are derived by calculating the relevant TP, FP, TN, and FN. TP: positive samples that are predicted by the model to be true classes. TN: negative samples that are predicted by the model to be in the negative class. FP: negative samples that are predicted by the model to be false classes. FN: positive samples predicted by the model to be false classes.

The precision formula is as follows:

$$\text{Precison} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{12}$$

Higher precision means that the changed regions are detected more accurately.

The recall formula is as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TN} + \text{TP}}. \tag{13}$$

Recall known as the full rate of detection is specific to the original sample. It denotes the probability of a positive sample being predicted from an actual true sample. The higher the recall is, the better the performance in detecting the true changed areas.

The formula for $F1$ is as follows:

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precison} + \text{recall}}. \tag{14}$$

$F1$ is defined as the summed average of precision and recall. $F1$ takes values between 0 and 1, with 1 meaning the best model output and 0 meaning the worst model output.

The formula for IoU is as follows:

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{15}$$

IoU measures the accuracy of detecting the corresponding object in a given dataset, and a higher value denotes a higher probability of detecting the true changed areas.

The formula for kappa is as follows:

$$\text{Kappa} = \frac{p_0 - p_e}{1 - p_e}. \tag{16}$$

Kappa is used for consistency testing. It can also be used as a measure of classification accuracy. The result is calculated as -1 to 1; however, kappa usually falls in the range of 0–1. p_e: the true value of each class in the confusion matrix is multiplied by the predicted value of each class and then divided by the square of the total sample size. p_0: the number of correctly classified samples in the confusion matrix (diagonal values) divided by the number of all samples.

D. Ablation Experiments

The baseline method for the ablation experiment is set up to detect directly in the transformer whether the increased or decreased area in the multitemporal RS images is the target changed area, such as changed areas for buildings, roads, or trees. The proposed method in this article has two main additional modules compared with the baseline method. The first module is a TMM module that is added to effectively exploit the spatial and temporal properties of multitemporal RS image datasets. The second module is a CDDM model that is added to better learn information from different maps and increase the overall network learning performance. Table I compares the performance of the algorithm before and after adding the two modules. Figs. 7 and 8 show some of the detection results of the proposed method on the two datasets.

Our model is proven to reduce the probability of pseudochanges appearing and improve the detection accuracy on both datasets. In Fig. 7, we find that both the changed areas of dense buildings [Fig. 7(a7)–(d7)] and the changed areas of sparse buildings [Fig. 7(a3)–(d3) and (a5)–(d5)] can be accurately detected. Fig. 7(a2)–(d2) and (a4)–(d4) shows that the proposed method is very effective for unchanged images in which the preimages/postimages change with the seasons. It also suppresses false detections well, reducing the probability of pseudochanges and improving detection accuracy. In Fig. 8, we examined the ground surface changes in several major cities in China. We understand that not only are many buildings detected well but also new foundations [Fig. 8(a1)–(d1)] and new roads [Fig. 8(a6)–(d6)]. The algorithm proposed in this article reduces the pseudovariation and class imbalance problems by comprehensively analyzing two publicly available real-world datasets.

1) Effect of TMM: As Table I shows, the $F1$ score improves from 92.20% to 95.04% (+2.84%) with the addition of TMM, and the precision and recall rates are improved by 1.11% and 4.58%, respectively. The algorithm speed is affected by the introduction of the TMM, which increases the processing time by 0.0021 s per image.

Nevertheless, the impact of the speed is at an acceptable level compared with the increase in precision and recall, as shown in Table I. To better illustrate the effectiveness of adding the module, this section also compares the detection effects under various changed regions. Five irregularly varying regions of visually changed feature regions are engaged in the experiment (on data LEVIR-CD). As Figs. 9 and 10 show, feature 1 is a multitemporal RS image pair with one of the pseudochanges; that is, it has unchanged areas. Feature 2 refers to changed areas for buildings with larger and sparser footprints. Feature 3 refers to changed regions where the building has a small footprint and is also very dense. Feature 4 refers to changed areas where a variety of buildings occur. Feature 5 refers to the changed areas in the buildings that are distributed throughout all areas. The comparison between

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TABLE I
ABLATION EXPERIMENTS BASED ON MODULES

Method	*Precision*	*Recall*	*F1*	*IoU*	*speed/s*
Baseline Method	92.80	91.61	92.20	92.65	**0.1504**
Baseline Method + TMM	93.91	96.19	95.04	93.62	0.1525
Baseline Method + TMM +CDDM	**97.05**	**97.79**	**97.41**	**96.92**	0.1699

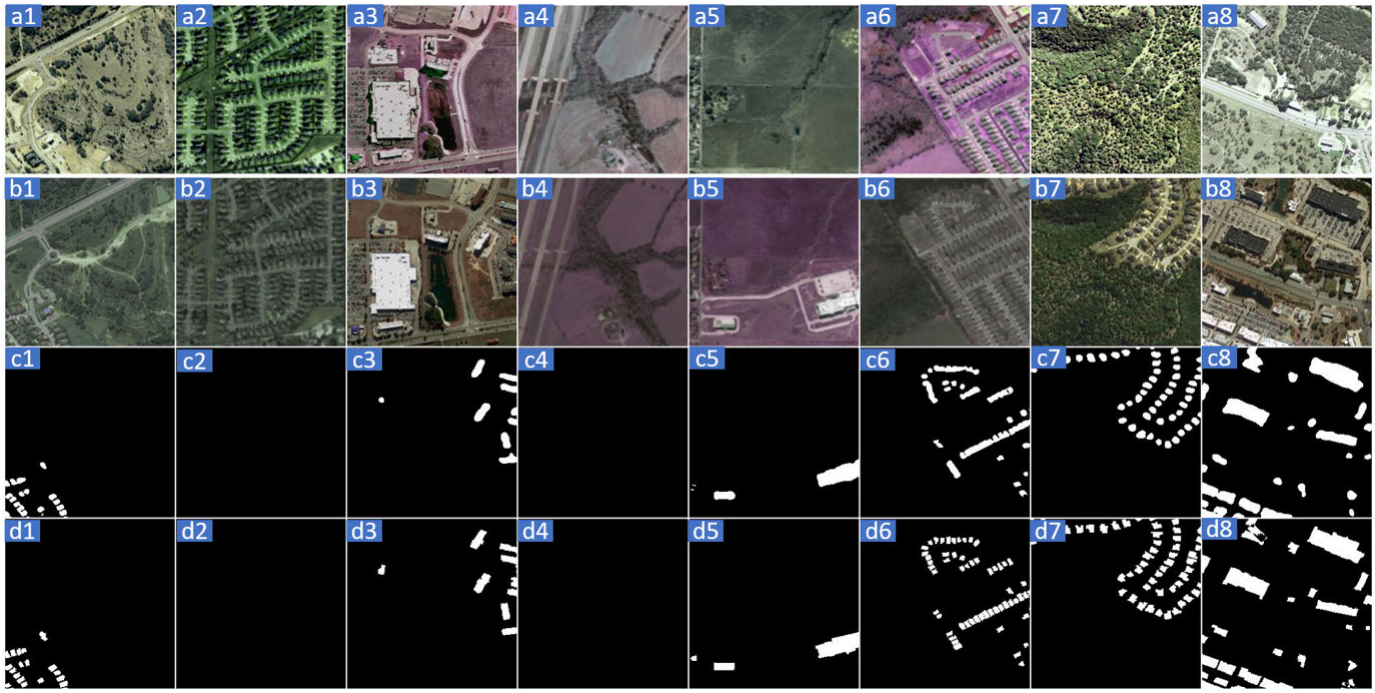


Fig. 7. Some of the detection results based on the proposed method (LEVIR-CD dataset): (a1)–(a8) prechange image, (b1)–(b8) postchange image, (c1)–(c8) ground truth, and (d1)–(d8) proposed method.

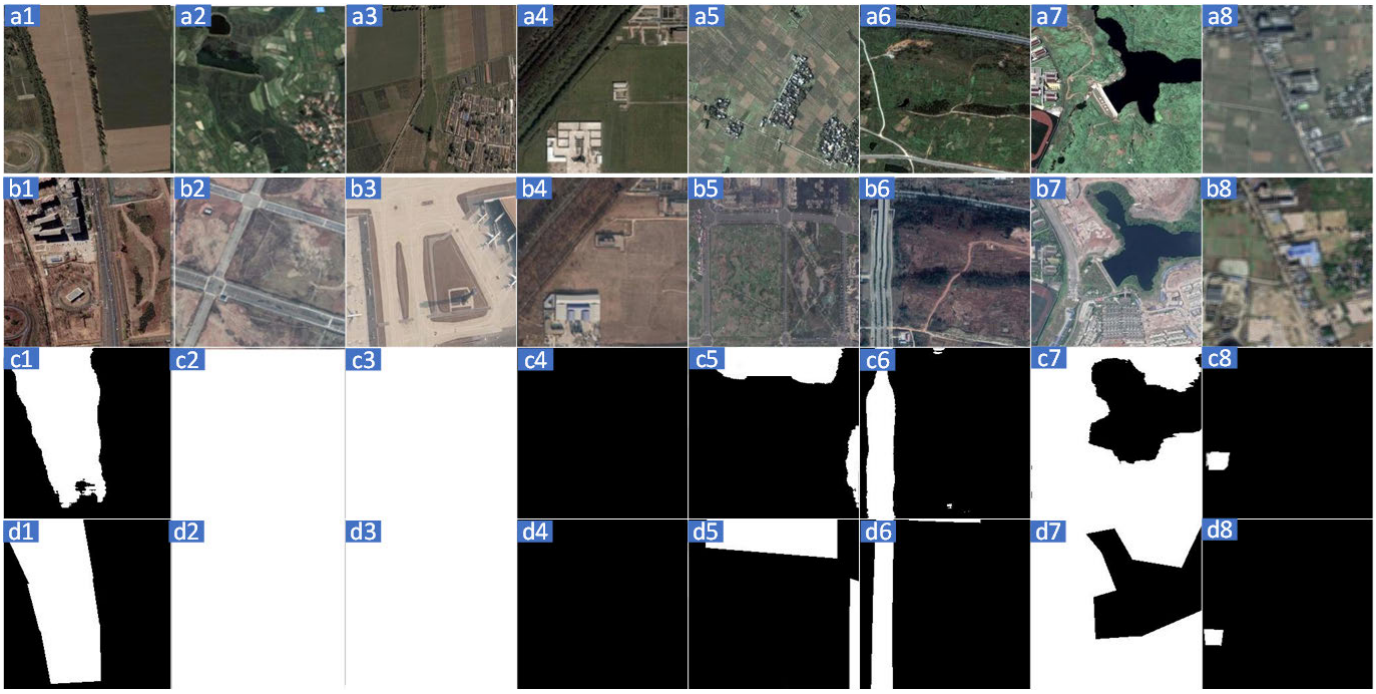


Fig. 8. Some of the detection results based on the proposed method (DSIFN-CD dataset): (a1)–(a8) prechange image, (b1)–(b8) postchange image, (c1)–(c8) ground truth, and (d1)–(d8) proposed method.

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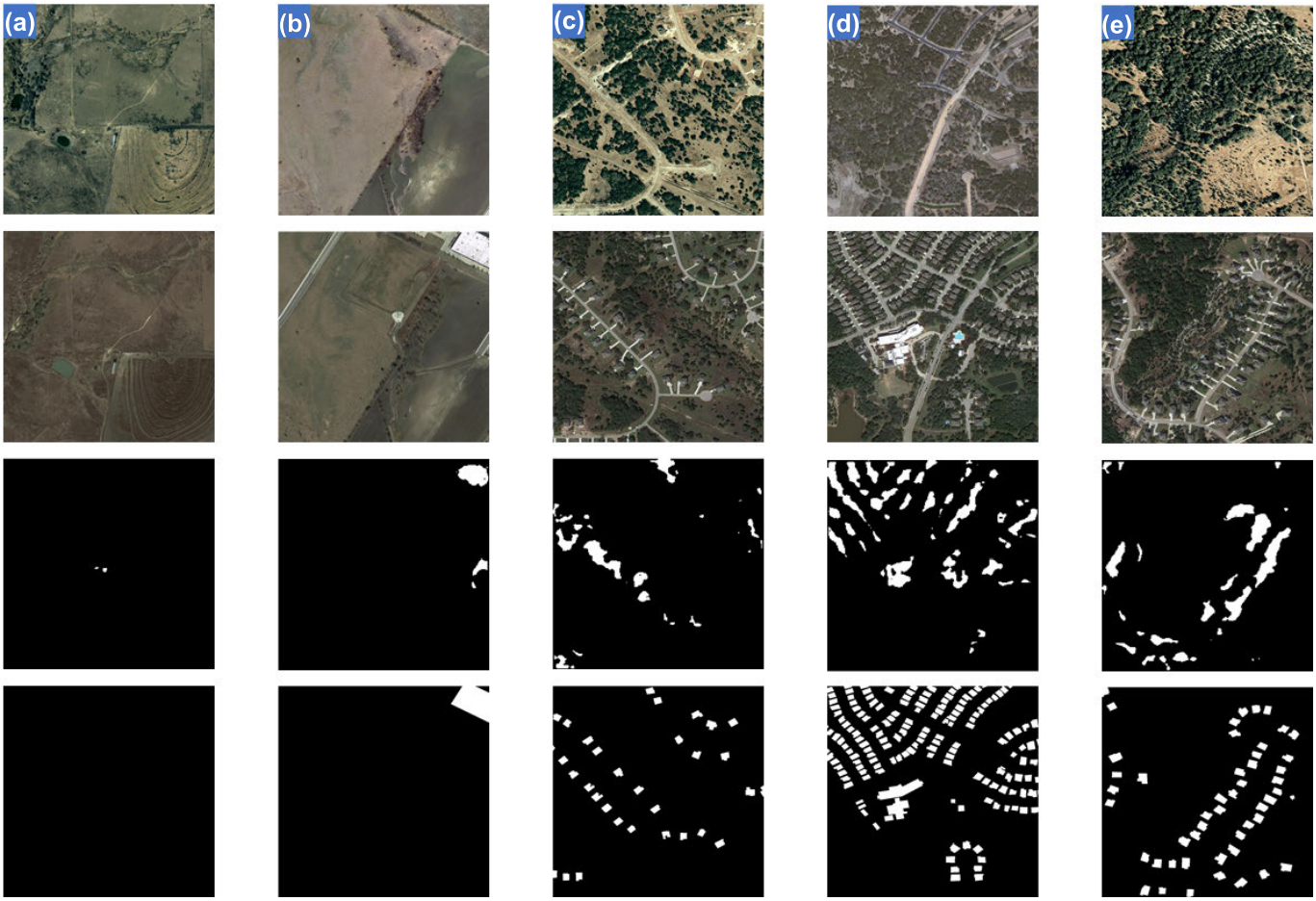


Fig. 9. Baseline method change detection results for different visual features: (a) feature 1, (b) feature 2, (c) feature 3, (d) feature 4, and (e) feature 5 (the first row of prechange images, the second row of postchange images, the third row of detection results, and the fourth row of ground-truth images).

Tables II and III shows that our work achieves better results than the baseline method by applying the TMM to the CD of high-resolution RS images. Notably, in feature 1, both the additional module and the baseline method reduce false CD detections.

However, in both Tables II and III, the additional module improves precision from 26.49% to 46.28% (+19.79%), the recall from 22.10% to 87.63% (+65.53%), and the kappa from 17.51% to 52.31% (34.8%). In Tables II and III, we observe that the TMM of this article effectively improves the CD results in each evaluation metric. We also find from Figs. 9 and 10 that the TMM is preferable to the baseline approach, whether it is better at reducing false detections in feature 1 where there are unchanged regions or at detecting changed areas where dense buildings are present. In total, the better TMM performance in this article compared with the baseline method illustrates the effectiveness of the TMM.

We can see in Table III that the TMM behaves differently with each feature; for example, it remains less effective in features 3 and 5, and we find from Fig. 10 that features 3 and 5 are dense buildings as changed areas that we need to detect. For this problem, we studied the difference map.

2) Effect of CDDM: As Table I shows, we add the CDDM to learn the difference map because of the large amount of useful change information contained in the difference map. We note the improvements of 4.25%, 6.18%, 5.21%, and 4.27% in precision, recall, $F1$, and IoU, respectively, compared with the baseline method. In addition, adding the CDDM is more successful in detecting changed regions than the TMM, with precision, recall, $F1$, and IoU improving by 3.14%, 1.6%, 2.37%, and 3.3%, respectively. Run time increased by only 0.0174 s.

Table IV and Fig. 11 demonstrate the increase in the overall performance of the method with the addition of the CDDM, requiring only minimal additional effort. In particular, the improvement in precision is especially noticeable.

To verify the validity of the CDDM in this article, we compare the two CD methods using the different modules in terms of performance. The methods involved in the comparison are based on the SA difference module CD method [44] and the multimethod stacked difference module CD method [45]. The first method utilizes a difference enhancement module consisting of two convolutional units and an SA module. The comparative results are shown in Table V: the SA difference module has a higher recall, but precision and IoU are 2.17% and 3.21% lower than those of our model, respectively. More importantly, the $F1$ of the two compared methods is lower than that of this article.

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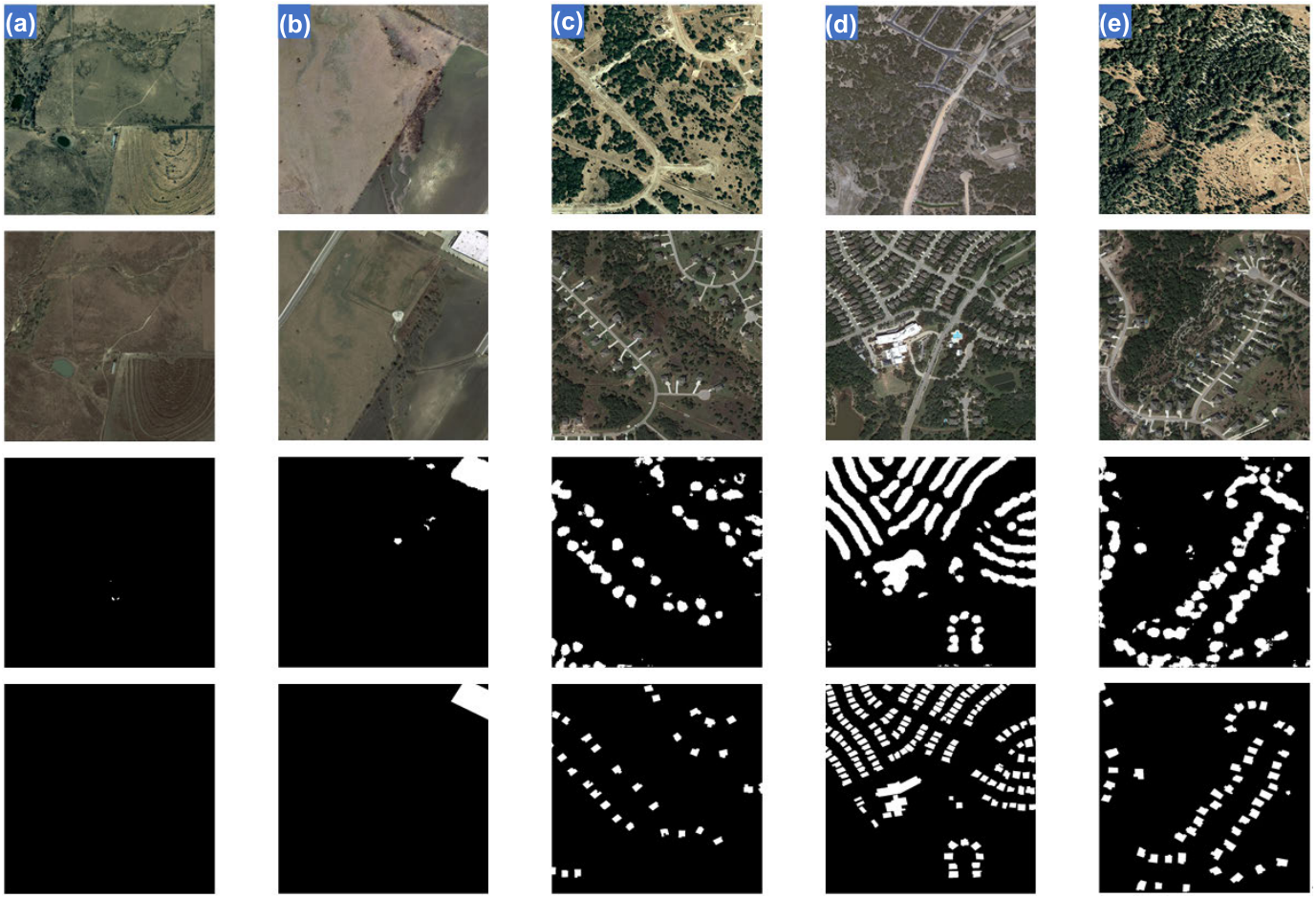


Fig. 10. TMM change detection results for different visual features: (a) feature 1, (b) feature 2, (c) feature 3, (d) feature 4, and (e) feature 5 (the first row of prechange images, the second row of postchange images, the third row of detection results, and the fourth row of ground-truth images).

TABLE II

COMPARISON OF PERFORMANCE OF BASELINE METHODS WITH DIFFERENT VARIATION CHARACTERISTICS (DATASET: LEVIR-CD)

Name	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	$Precision(\%)$	$Recall(\%)$	$F1(\%)$	$IoU(\%)$	$Kappa(\%)$
Figure1	✓					98.15	97.98	98.56	98.65	98.15
Figure2		✓				77.46	53.31	63.16	46.15	62.45
Figure3			✓			13.46	13.31	13.36	11.71	12.16
Figure4				✓		57.47	32.67	41.67	26.31	30.98
Figure5					✓	26.49	22.1	24.08	13.69	17.51

TABLE III

COMPARISON OF PERFORMANCE OF TMM WITH DIFFERENT VARIATION CHARACTERISTICS (DATASET: LEVIR-CD)

Name	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	$Precision(\%)$	$Recall(\%)$	$F1(\%)$	$IoU(\%)$	$Kappa(\%)$
Figure1	✓					98.16	98.98	99.28	99.65	98.23
Figure2		✓				83.54	97.82	90.01	82.11	88.81
Figure3			✓			38.88	90.61	54.41	37.37	48.94
Figure4				✓		66.26	91.19	77.01	62.61	63.45
Figure5					✓	46.28	87.63	60.57	43.44	52.31

3) Comparison of Loss Functions and Precision: The precision and loss curves for training and validating different algorithms are shown in Fig. 12. All the methods evaluated are convergent, but our method is smoother and more stable, as shown by zoomed-in image [Fig. 12(a1)–(c1)]. There is no significant difference between the training performance and verification performance.

From the precision curves [Fig. 12(a2)–(c2)], the precision of these types of methods improved during training and validation. We discover that the validation curves of the algorithms in this article are much better trained than those of the other algorithms. In addition, all training curves improve. However, our training curves increase faster than other transformer-based algorithms.

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TABLE IV
COMPARISON OF PERFORMANCE OF CDDM WITH DIFFERENT VARIATION CHARACTERISTICS (DATASET: LEVIR-CD)

Name	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	$Precision(\%)$	$Recall(\%)$	$F1(\%)$	$IoU(\%)$	$Kappa(\%)$
Figure1	✓					99.98	99.99	99.99	99.98	99.99
Figure2		✓				99.43	98.65	95.13	90.71	94.54
Figure3			✓			70.69	86.24	77.7	63.53	70.71
Figure4				✓		83.31	82.43	82.87	70.48	71.08
Figure5					✓	78.61	90.12	81.79	69.21	74.12

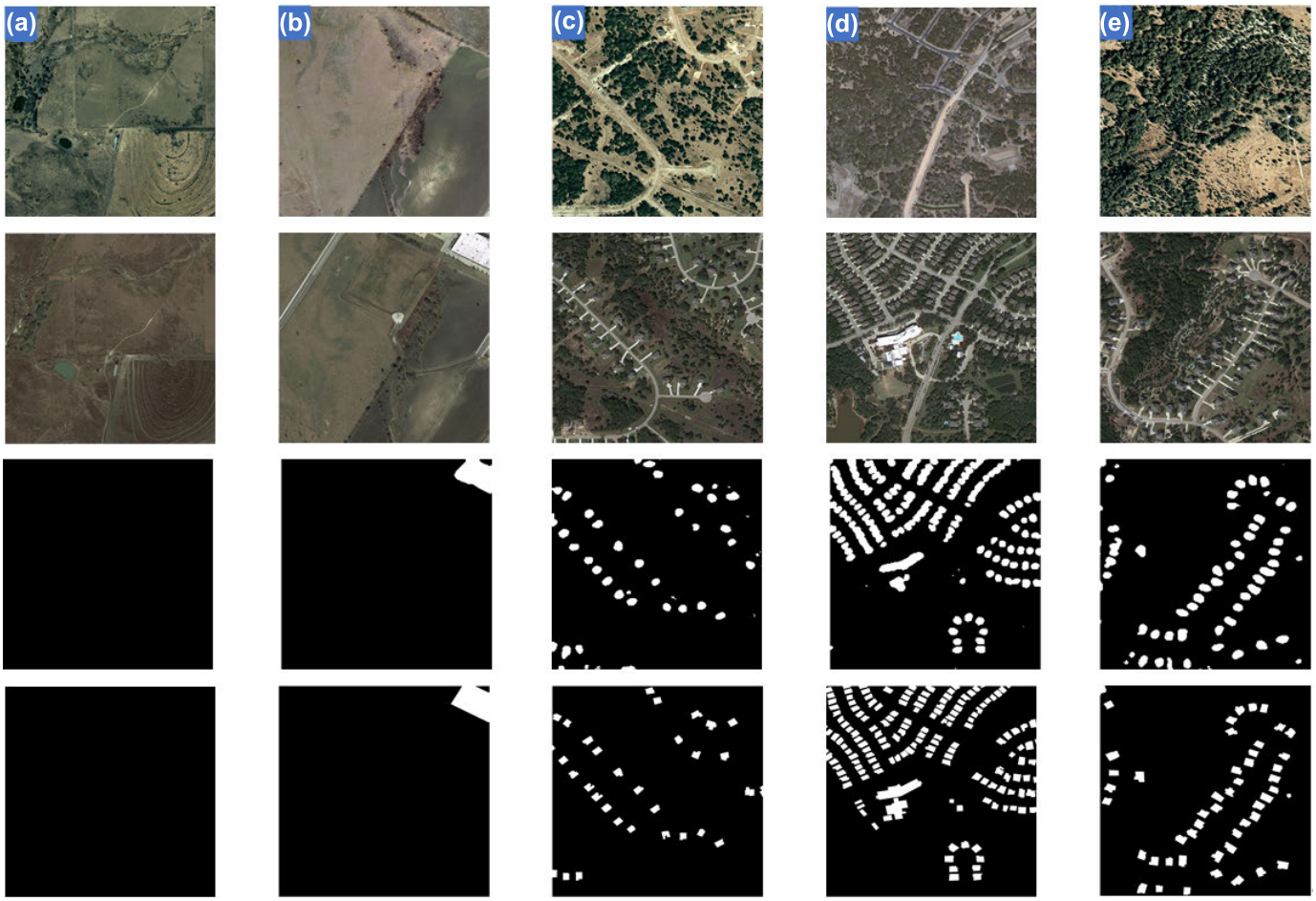


Fig. 11. CDDM module change detection results for different visual features: (a) feature 1, (b) feature 2, (c) feature 3, (d) feature 4, and (e) feature 5 (the first row of prechange images, the second row of postchange images, the third row of detection results, and the fourth row of ground-truth images).

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT MODULES (DATASETS: LEVIR-CD AND DSIFN-CD)

Extraction method	$Precision$ (%)	$Recall$ (%)	$F1$ (%)	IoU (%)
SA difference module	96.54	**98.95**	97.73	93.28
Multi-method stacked difference	91.31	88.69	89.91	83.69
CDDM	**98.71**	98.73	**98.72**	**96.49**

E. Comparative Experiments

Two different types of datasets are used for the experiments. LEVIR-CD data detect changes for all kinds of changed buildings, and DSIFN-CD data detect changed areas where the ground surface changes (not only for buildings). The approaches are compared not only with the transformer as the backbone network but also with convolution as the backbone network. This is done to ensure a reliable and fair comparison. In the following content, each network is documented in detail.

1) *Siamese Network and NestedUNet (SNUNet) [46]:* A densely linked network for change detection: SNUNet-CD (combination of a Siamese network and NestedUNet). A densely connected Siamese network inspired by DenseNet and NestedUNet was designed for CD. It maintains a high-resolution and fine-grained representation through dense skip connections between the encoder and the decoder and between the decoder and the decoder.

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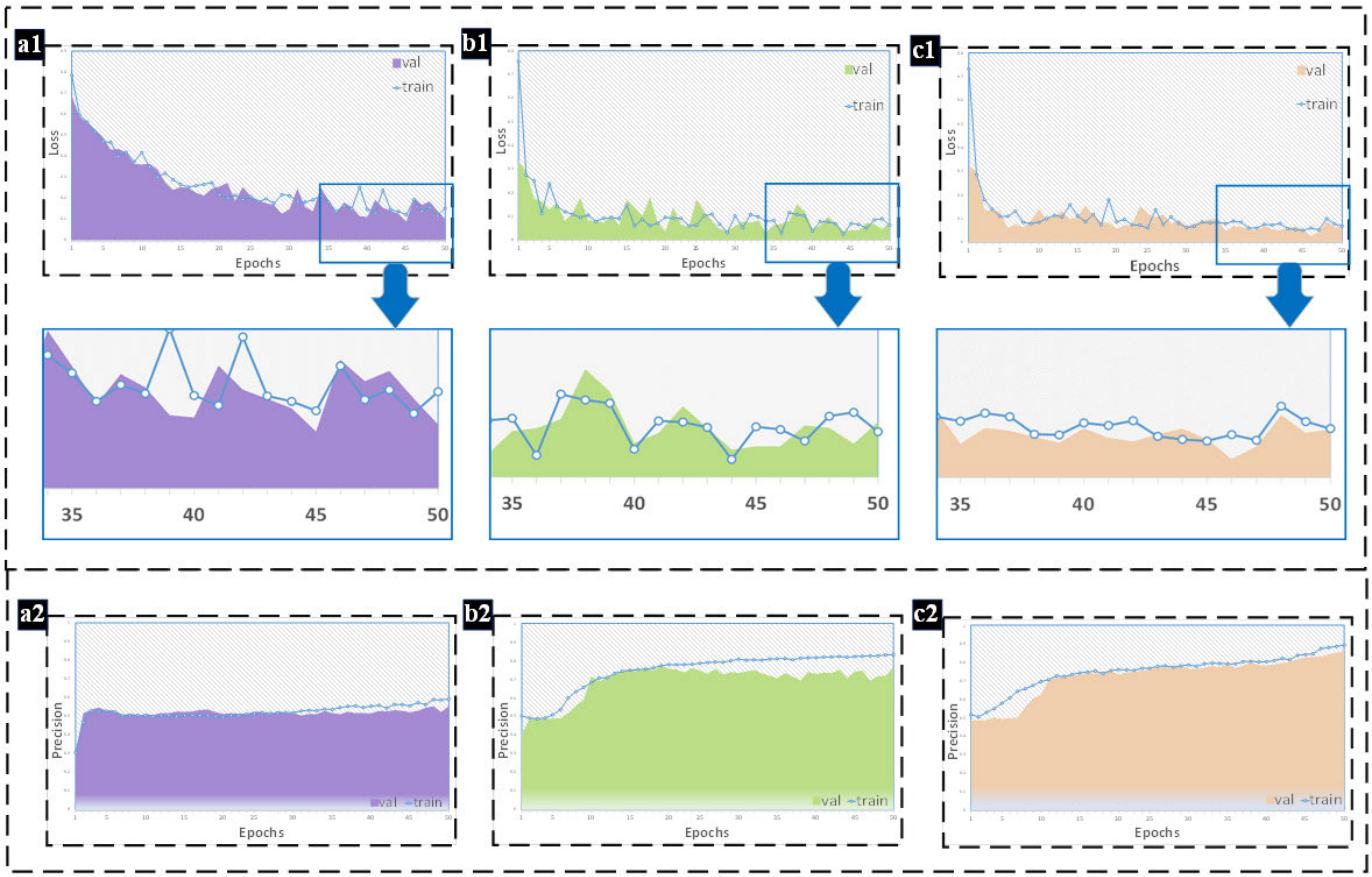


Fig. 12. Training and validation loss curves for different models trained on the LEVIR-CD dataset: (a1)–(c1) BIT [31], [45], T-C, and the proposed STADE-CDNet. Training and validation precision curves for different models trained on the LEVIR-CD dataset: (a2)–(c2) BIT, T-C, and the proposed STADE-CDNet.

2) *FC-Siam-Di [47]:* This is a method for obtaining details in shallow network information by using the skip connection concept of UNet. These details supplement the more abstract and globalized information after encoding. This better produces accurate class predictions with precise boundaries in the output image. The network detects the details by combining two skip connections in the decoding step.
3) *FC-Siam-Conc [47]:* This feature stitching network uses a UNet to stitch CD features and then extract various features from the CD images. The final CD result is obtained.
4) *Iterative Difference-Enhanced Transformer (IDET) [48]:* IDET is a novel module called iterative difference-enhanced transformers (IDETs) for CD. IDET optimizes the feature difference to highlight changes and suppress unchanged regions.
5) *Tiny [49]:* Tiny is a lightweight and effective change detection model. It utilizes a Siamese UNet architecture and introduces the mix and attention mask block (MAMB) for feature fusion and spatial-semantic attention.
6) *Fully Transformer Network for Change Detection of Remote Sensing Images (FTN) [50]:* FTN framework improves RSI-CD by leveraging transformers for global feature extraction and a pyramid structure for multilevel feature aggregation. The progressive attention module enhances feature representation, while deeply supervised learning with boundary-aware loss functions further enhances training.
7) *Dual Task Constrained Deep Siamese Convolutional Network (DTCDSCN) [51]:* This is a twin convolutional neural network with dual task constraints that consists of three subnetworks: a CD network and two semantic segmentation networks (SSNs).
8) *Bitemporal Image Transformer (BIT) [31]:* This structure is based on a combination of convolutions and a transformer. Convolutions are used to extract high-level semantic information in the CD difference map. Then, encoding and decoding the transformer network are used to extract regions of interest in the CD difference map.
9) *Transformer-Based Siamese Network for Change Detection (T-C) [45]:* This is a network for RSI-CD using a transformer. The semantic information in the CD areas is enriched by dividing network encoding into four parts. Ultimately, the desired change information is detected.

In Figs. 13 and 14, the qualitative detection results for the changed areas on both datasets are shown for each algorithm. Tables VI and VII show the quantitative detection results.

Some of the detection results are shown in Fig. 13. Three different types of test image pairs are shown here (the first row is the first pair of multitemporal RS images, the second row is the second pair of multitemporal RS images, and the third

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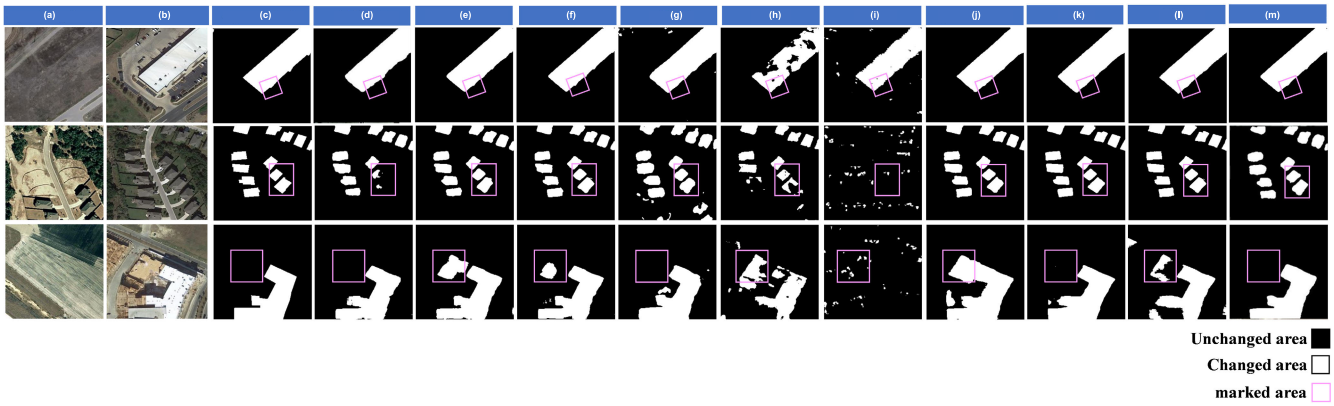


Fig. 13. Qualitative results of different CD methods on LEVIR-CD data: (a) prechange image, (b) postchange image, (c) ground truth, (d) SNUNet, (e) FC-Siam-Di, (f) FC-Siam-Conc, (g) IDET, (h) Tiny, (i) FTN, (j) DTCDSCN, (k) BIT, (l) T-C, and (m) proposed method.

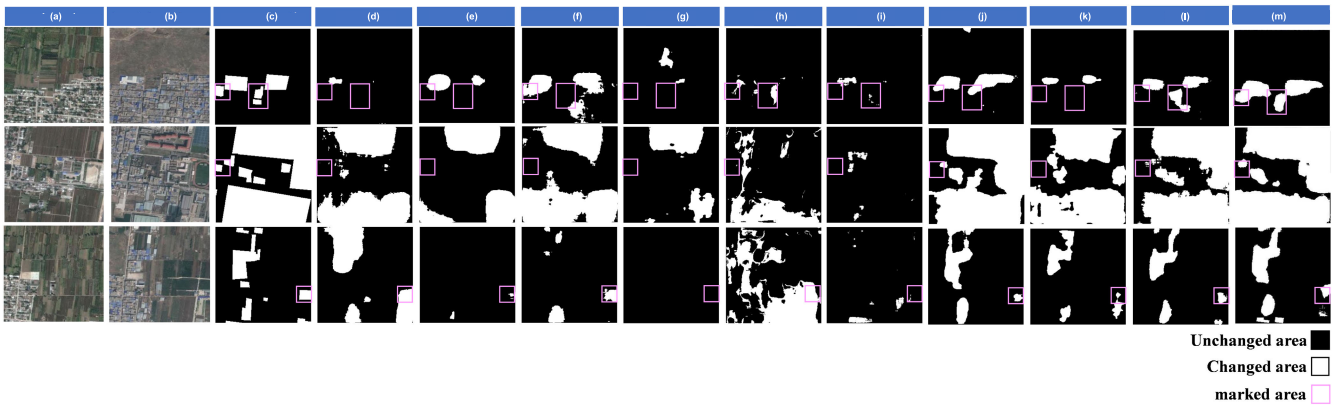


Fig. 14. Qualitative results of different CD methods on DSIFN-CD data: (a) prechange image, (b) postchange image, (c) ground truth, (d) SNUNet, (e) FC-Siam-Di, (f) FC-Siam-Conc, (g) IDET, (h) Tiny, (i) FTN, (j) DTCDSCN, (k) BIT, (l) T-C, and (m) proposed method.

TABLE VI
PERFORMANCE COMPARISON WITH OTHER METHODS (DATASET: LEVIR-CD)

Method	Precision (%)	Recall (%)	F1 (%)	IoU (%)
SNUNet [46]	78.91	72.58	75.61	74.33
FC-Siam-Di [47]	85.41	86.27	85.83	83.48
FC-Siam-Conc [47]	**90.67**	85.26	87.88	75.67
IDET [48]	88.32	83.78	85.87	72.64
Tiny [49]	88.43	68.31	76.71	62.21
FTN [50]	40.12	39.12	36.14	25.74
DTCDSCN [51]	86.12	84.71	85.4	81.32
BIT [31]	90.27	**88.65**	**89.45**	**88.11**
T-C [45]	**91.56**	**89.78**	**90.66**	**83.69**
Proposed	**92.79**	**92.47**	**92.63**	**93.81**

* Color convention: **best**, **$2^n d$-best**, **3rd-best**.

row is the third pair of multitemporal RS images). Our method visually detects changed regions better than other methods, both for large and sparse building groups and for small and dense buildings. For example, in the first row of the CD image pair, our method is much better at detecting the detailed parts of the changed building than the other nine comparison methods. This reduces missed detections and improves CD accuracy. In the second row of CD image pairs, our approach with T-C is more effective in detecting changed dense building regions than the others while ensuring the structural integrity of the detected buildings. In the third row of the CD image pair, our method not only accurately detects changed building areas compared with the baseline approaches but also reduces the number of missed detections. As illustrated above, the method proposed in this article is superior to the other nine algorithms.

Some of the detection results are shown in Fig. 14, and three different types of test image pairs are shown here. In this DSIFN-CD dataset, we detect changed areas not only for buildings but also for all ground surface changed areas that occur at different times. In the first row, our algorithm can detect the changed areas better than other baseline algorithms during the detection process, which reduces the missed and false detection regions and improves detection precision.

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TABLE VII
PERFORMANCE COMPARISON WITH OTHER METHODS (DATASET: DSIFN-CD)

Method	Precision (%)	Recall (%)	F1 (%)	IoU (%)
SNUNet [46]	62.31	47.64	53.99	40.67
FC-Siam-Di [47]	**71.89**	73.48	72.68	45.65
FC-Siam-Conc [47]	**69.35**	58.47	63.45	43.86
IDET [48]	65.22	56.43	59.37	39.64
Tiny [49]	63.42	36.65	46.56	30.34
FTN [50]	70.12	51.21	59.09	43.11
DTCDSCN [51]	59.81	74.93	62.95	45.31
BIT [31]	71.19	**75.85**	**80.75**	**59.48**
T-C [45]	**90.64**	**86.67**	**88.61**	**78.42**
Proposed	**93.51**	**88.13**	**90.74**	**79.05**

* Color convention: **best**, $2^n d$-**best**, 3rd-**best**.

In summary, from both Figs. 13 and 14, the algorithm proposed in this article is better than the other comparison algorithms.

We also conducted sufficient experiments on the LEVIR-CD dataset. As Table VI shows, the network proposed in this article achieves the best in all evaluation metrics, and the IoU increases by 10.12% over T-C. In comparing Fig. 13, we find that the robustness of the STADE-CD network outperforms that of the feature-based approach in complex scenarios.

Moreover, we conduct sufficient experiments on the DSIFN-CD dataset. As Table VII shows, our algorithm is 21.62%, 13.2%, and 33.4% higher in precision, recall, and IoU, respectively, than those of UNet-like networks. Overall, the detection result of our approach is optimal. In summary, comparison experiments were used to evaluate the performance of STADE-CDNet. We discovered that our research $F1$ rates are at least 2.1% and 1.97% higher than those of other studies in the DSIFN-CD and LEDVIR-CD datasets. Our method in Table VI is better than the other methods. In Table VII, we found that the quantitative metrics are higher than those of the T-C methods. Therefore, our method is much better than other recent methods in terms of quantitative metrics.

In this study, we compare the algorithm running times of various state-of-the-art models, including SNUNet, FC-Siam-Di, FC-Siam-Conc, IDET, Tiny, FTN, DTCDSCN, BIT, T-C, and the proposed method. The algorithm running times are measured for processing two batches of data, as shown in Table VIII. Analyzing the approach complexity, we observe variations in the running times of different models. SNUNet, FC-Siam-Di, FC-Siam-Conc, and the proposed method exhibit relatively shorter running times, ranging from 32.98 to 67.45 s in LEVIR-CD data. This suggests that these models have lower computational complexity and are more efficient in processing the data.

On the other hand, IDET, Tiny, FTN, and DTCDSCN demonstrate comparatively longer running times in two data, ranging from 80.24 to 300 s in LEVIR-CD data. These models may have higher computational complexity, requiring more time for computation and processing. It is worth noting that the proposed method showcases a significantly shorter running time compared with most of the state-of-the-art models, indicating its potential for improved efficiency in CD tasks.

In analyzing algorithm complexity, the algorithm complexity evaluation reveals a trade-off between computation time and model performance. While some models achieve faster processing times, others may provide more accurate results but require additional computational resources.

F. Discussion

In this research, we conducted extensive experiments on various datasets to test our method and draw conclusions. The proposed method exploits the fact that high-resolution RSI-CD datasets typically consist of spectral vectors in both temporal and spatial dimensions. We extracted corresponding information of the changed regions by utilizing temporal and spatial information in the STADE-CD network. In addition, we explored a CDDM to extract effective change information from difference maps. The experimental results demonstrate the superiority of our approach over baseline methods, as illustrated in Figs. 13 and 14. Furthermore, our method achieves the best precision and recall rates on all datasets. Notably, balancing the time and spatial information in the transformer network is challenging. Our proposed method not only learns useful information from the temporal dimension but also performs better than other time-series methods. This indicates that our model is more stable and performs better.

Moreover, we study CDDM, because it is a crucial step in CD. We observe that when algorithms involve many calculation processes, directly using subtraction results in poor detection accuracy [31]. The CDDM we proposed can effectively obtain change information in multitemporal RS image pairs and reduce missed and false detections. Moreover, incorporating the CDDM into the network for learning enables the module to learn more change information and deeper information, thereby improving the overall network performance. The ablation experimental results in Table I demonstrate that adding the CDDM only increased the runtime of each image by 0.0174 s. In addition, Tables VI and VII show that the CDDM improves the overall qualitative indicators, highlighting the effectiveness of the CDDM.

We further investigate using temporal change information in time series and its application to extract valid information from RS multitemporal image pairs. Our initial approach employs the TMM to detect changed areas between the first and last period images without effectively utilizing other change information. However, this approach is not efficient, as even deep learning frameworks struggle to extract irregularly changed ground surfaces at different times.

Through extensive experiments, we propose an approach that improves performance by using transformers to extract

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TABLE VIII
EXECUTION TIME (IN SECONDS) OF ONE EPOCH TRAINING OF DIFFERENT METHODS (BATCH SIZE = 2 AND ONE ITERATION)

Data set	SNUNet [46]	FC-Siam-Di [47]	FC-Siam-Conc [47]	IDET [48]	Tiny [49]	FTN [50]	DTCDSCN [51]	BIT [31]	T-C [45]	Proposed
LEVIR-CD	67.45	45.19	48.3	167.78	267.55	300.87	80.24	30.56	46.07	32.98
DSIFN-CD	509.48	479.42	406.31	749.33	1014.26	960.65	539.36	280.71	363.16	301.63

semantic information from high to low levels in space, followed by extracting temporal information from the image at different times using TMM. These steps provide meaningful information without relying on independent temporal change vectors. This result is consistent with previous CD algorithms that combine LSTM with UNet networks [36].

Unfortunately, we cannot determine from these data that we can solve many CD problems using the STADE-CD network model. One limitation is that our method is not suitable for more diverse datasets. Despite its preliminary character, this study indicates that our method is valid.

Thus, we believe that the experimental results will provide valuable feedback for future improvements in extracting change information from high-resolution RS images and detecting directly in the transformer whether the increased or decreased area in the multitemporal RS images is the target changed area, such as changed areas for buildings, roads, or trees.

IV. CONCLUSION

In this article, we propose the STADE-CDNet deep learning method that addresses the pseudochanges and inherent imbalance challenges in RS change detection datasets by leveraging the characteristic that the CD datasets consist of spectral vectors in both temporal and spatial dimensions. Specifically, we combine a transformer with the TMM to extract spatial semantic and temporal feature information. In addition, by placing the TMM after transformer encoding, this framework not only alleviates the vanishing gradient during backpropagation but also creates effective feature representations by extracting spatial and temporal information during learning. In addition, we developed a CDDM to improve the network's learning ability by learning the difference map in the CD, which detects changed areas better. Qualitative and quantitative analyses prove that our method greatly enhances the probability of true positives and significantly improves the precision and recall of all datasets. We still have problems detecting insufficient semantic information and complex module design in the design of the TTMM. In the future, we will adopt the network search method based on reinforcement learning to further improve the effectiveness of the network model design. For the changed multivariable target contour, we will further explore the time information of CD to improve the change detection accuracy.

ACKNOWLEDGMENT

The authors would like to thank the editor who gave constructive comments and helped this article's quality. The same appreciation goes to the published code of Wele Gedara Chaminda Bandara's models, Hao Chen's model, Fang's model, Rodrigo Caye Daudt's models, and Liu's model for comparison.

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